

Immunobiology of endometriosis

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Objective: To provide a review of the humoral and cellular immunology of endometriosis and to discuss the rationale for future approaches to diagnosis and treatment.

Design: Literature survey.

Result(s): Defective immunosurveillance in women who are destined to develop endometriosis may allow for the survival of ectopic endometrial tissue. The evidence includes endometrial cell resistance to apoptosis, perhaps through the secretion of proteins that interfere with implant recognition and/or FasL expression by stromal cells, inducing apoptosis of Fas-bearing immune cells. Although the immune response may be defective, aspects of it clearly are enhanced in endometriosis, as is seen by the generalized polyclonal B-cell autoimmune activation and secretion of immune proteins. Several cytokines, chemokines, and growth factors (including vascular growth factors) are increased in women with endometriosis.

Conclusion(s): A complex network of locally produced cytokines modulate the growth and inflammatory behavior of ectopic endometrial implants. Proinflammatory proteins from endometriotic lesions and associated immune cells contribute to the enhanced inflammatory reaction associated with endometriosis that subserves the survival of these lesions instead of leading to their demise. (*Fertil Steril*® 2001;75:1–10. ©2001 by American Society for Reproductive Medicine.)

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Endometriosis is a common gynecological disorder, affecting at least 10% of reproductive-aged women. It is characterized by the growth of endometrial tissue outside the uterine cavity. The association of endometriosis with pelvic pain and impaired fecundity has been recognized since the 1920s (1). The effect of endometriosis on the success of assisted reproductive techniques is controversial. However, some studies suggest that the presence of this clinical condition has a negative impact on implantation and pregnancy rates even when fertilization occurs extracorporeally (2, 3).

Current therapies for treating endometriosis include surgical and medical techniques aimed at cytoreduction or cytosuppression, respectively, of extant implants. Because endometriosis has a tendency to progress or recur following surgery (4) or after cessation of medical suppression (5), these strategies serve only to temporize symptom relief. The elucidation of mechanisms that underlie endometriosis pathophysiology should lead to more effective therapies for this debilitating syndrome.

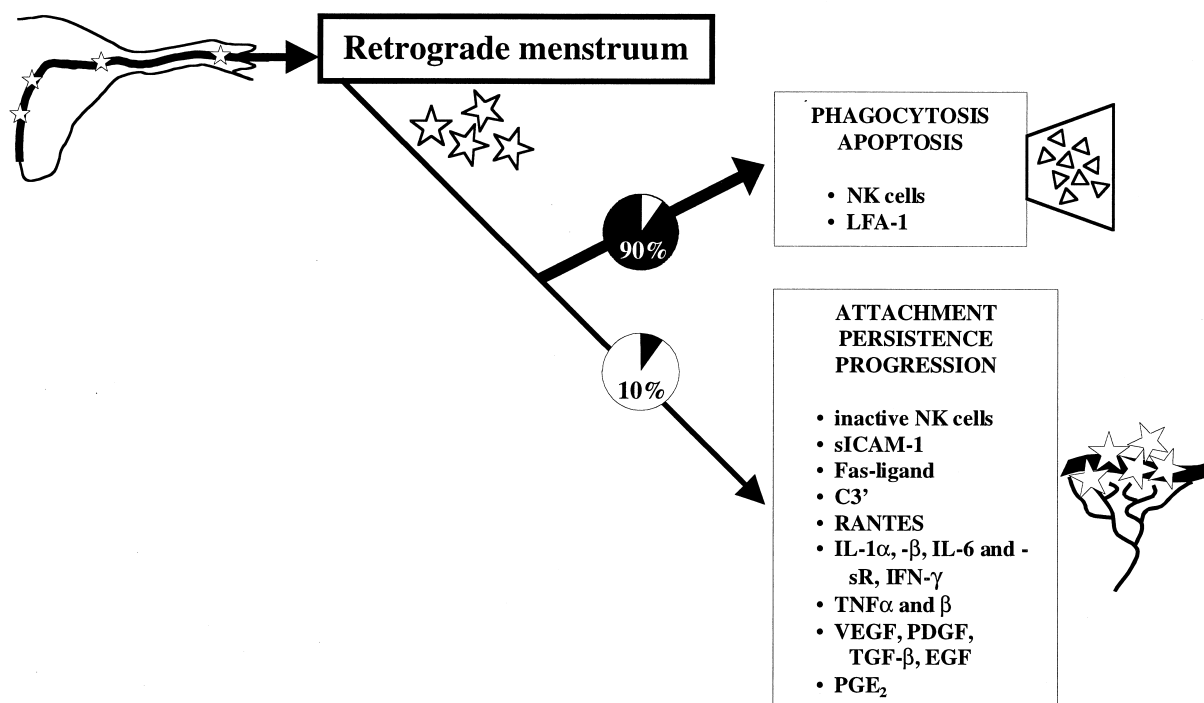
IMPLANTATION OF ENDOMETRIAL FRAGMENTS

Although multiple theories exist regarding the etiology of endometriosis, the implantation hypothesis of Sampson is the most widely accepted, albeit with some disagreement (6). Recent studies have supported the Sampson theory by documenting that retrograde menstruation (7) and peritoneal spillage of viable endometrial epithelial cells (8) occur frequently in cycling women. The attachment and adhesion of shed endometrial cells to peritoneal and subperitoneal surfaces are likely to involve the expression of extracellular membrane adhesion molecules and their co-receptors (9, 10).

Shed endometrial fragments are believed to accumulate in the dependent portions of the pelvis and adhere to the peritoneal surface. Microscopic defects may allow the cells to contact the submesothelial matrix (11), where they proliferate, spread, and grow, sometimes invading deeply into the subperitoneal space (12). Thus, accommodating factors must exist

FIGURE 1

The almost uniform phenomenon of peritoneal spillage of viable endometrial epithelial cells leads to phagocytosis or apoptosis in the majority of women (90%). The remaining fraction (10%) see the attachment, persistence and progression of the intraperitoneal fragments that define endometriotic lesions. Various inflammatory proteins are responsible for either fate, as listed in the figure and discussed in the text.



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that support the persistence of endometriotic lesions within the peritonea of the small subset of ovulatory women (10%–15%) who develop clinical evidence of this syndrome (Fig. 1). Other investigators and we have proposed that defective immunosurveillance in women who are destined to develop endometriosis may support the attachment, persistence, and progression of ectopic endometrial tissue in these women (13). Indeed, several studies indicate that the endometriosis syndrome is associated with immune-cell suppression.

Endometrial cells have been reported to be inherently resistant to apoptosis and phagocytosis (14), yet endometriosis occurs in only a minority of menstruating women (~10%–15%). The mechanisms by which regurgitated endometrial cells are cleared from the peritoneal cavity in the majority of women are poorly understood. However, it has been suggested that natural killer (NK) cells may subserve this role.

NK cells are the effector cells that usually recognize and destroy tumor cells, virus-infected host cells, and transplanted foreign cell lines. Oosterlynck and colleagues were the first to demonstrate decreased NK activity and cytotoxicity

against autologous endometrial cells in women with endometriosis, which correlated with the stage of the disease (15). The same group later showed that peritoneal fluid from women with endometriosis contained significantly greater NK cell suppressive activity than peritoneal fluid from fertile controls (16). Other investigators confirmed these findings in the serum (17) and pelvic fluid (18) of endometriosis patients.

One mechanism by which evasion of immunosurveillance might be accomplished is via the secretion of proteins that interfere with immunocyte–endometrial implant recognition. One such factor is the soluble form of the adhesion molecule ICAM (intercellular adhesion molecule)-1. ICAM-1 is a co-receptor for the cell-surface integrin ligand expressed on a variety of immune cells. Release of soluble ICAM-1 (sICAM-1) effectively generates a soluble receptor that binds leukocyte-associated ligands and competes for the ability of these leukocytes to participate in homotypic and heterotypic cell–cell interactions. Viganò and colleagues have shown that isolated endometriotic stromal cells secrete more sICAM-1 than matched stromal cells from eutopic endome-

trium. Furthermore, expression of ICAM-1 mRNA and protein is enhanced in endometriotic vs. endometrial stromal cells in culture (19).

Another hypothesis involves the Fas–Fas ligand (FasL) expression system. FasL-expressing cells induce apoptosis when they bind to Fas-bearing immune cells. Garcia-Velasco et al. showed that macrophage-conditioned media induced FasL expression by endometrial stromal cells, thus suggesting that peritoneal macrophages in endometriosis might stimulate a Fas-mediated apoptosis of immune cells as a further means of escape from immune surveillance (20).

HUMORAL IMMUNITY

Autoimmune Phenomena

Approximately 20 years ago, Weed and Arquembourg (21) observed IgG and complement deposits in the eutopic endometrium of women with endometriosis. These investigators hypothesized that an autoimmune mechanism might explain poor reproductive outcomes in some women with this syndrome. A decade later, Mathur et al. (22) were the first to describe in patients with endometriosis autoantibodies which recognize endometrial antigens (ranging in size from 34–140 kD) that are candidates for the autoantigens responsible for the immune response (23). Similar results have been reported elsewhere (24).

Wild et al. characterized one of the putative autoantibodies as a protein that cross-reacts with the enzyme carbonic anhydrase (25). It has been proposed that serum antibody titers to this or other autoantigens could be useful as a screening test for endometriosis risk. Using an endometrial carcinoma cell-based immunofluorescence assay, Wild et al. reported a sensitivity of 83% and a specificity of 65% among infertile patients with laparoscopy-proven endometriosis (25). Gleicher and colleagues (26) noted that a significant proportion of women with endometriosis (40%–60%) have elevated autoantibody titers when tested against a panel of autoantigens. Thus, in addition to the development of specific antiendometrial antibodies, a generalized polyclonal B-cell autoimmune activation is associated with some cases of endometriosis (27).

Dmowski et al. (28) evaluated the effect of autoantibodies on in vitro fertilization (IVF) success rates among women with endometriosis. In their retrospective analysis of 50 patients who had a history of endometriosis, those women with evidence of ≥ 3 autoantibodies (“positive”) had a significantly lower pregnancy rate per transfer (23% vs. 46%) than women with ≤ 2 autoantibodies (“negative”). It should be noted that 80% of the “antibody-positive” patients had evidence of active endometriosis, whereas only 36% of those classified as “antibody-negative” had active disease at the time of the IVF cycle.

In a smaller subset of these “antibody-positive” patients, 8 of 10 who were treated with 10–15 mg/d of prednisone

during the entire IVF cycle became pregnant, whereas none of the 25 “antibody-positive” endometriosis patients who were not treated with glucocorticoids conceived. This very preliminary report represents some of the only evidence suggesting that steroid therapy directed at the autoinflammation associated with endometriosis might be of clinical benefit with respect to infertility (see below).

Other components of the humoral immunity pathway appear to be enhanced in endometriosis, including the C3' component of the complement cascade (29) and fibronectin (30). Although some investigators have reported increased concentrations of prostaglandins in the cul-de-sac peritoneal fluid of patients with endometriosis (31), others have been unable to confirm these results (32). Menstrual-cycle effects and control-group selection do not completely explain the differences in these findings.

CELL-MEDIATED IMMUNITY

Leukocytes and Peritoneal Inflammation

A compelling body of evidence suggests a cellular immunological basis for endometriosis-associated infertility. Increased numbers of peritoneal macrophages and lymphocytes can be found in infertile women with endometriosis (33, 34). Some of these cells appear to be activated in patients with endometriosis (35–37).

Recent studies indicate that a variety of resident lymphoid cells can be identified histologically within endometriotic implants, and these cells are likely to contribute to the implants' pathophysiology (38). In addition to increased numbers of macrophages, women with advanced endometriosis also have increased numbers of pelvic fluid T-lymphocytes (39). As noted above, ECM molecules such as C3' could play a role in the proliferation of immune cells. Decreased NK-cell activity was observed in peritoneal fluid (16) and serum (17) from women with endometriosis.

Cytokines and Growth Factors Associated with Endometriosis

Cytokines and growth factors are proteins or glycoproteins typically secreted into the extracellular environment by leukocytes or other cells. In the extracellular environment, cytokines and growth factors exert their effects on the same (autocrine activity) or nearby cells (paracrine activity). Some of these proteins can circulate or traverse body cavities and thus may exert endocrine activities as well. In specific cases, cytokines exist in cell-membrane-associated forms where they act on adjacent cells (juxtacrine activity).

Cytokines are key mediators of intercellular communication within the immune system. They act in a pleiotropic fashion on a variety of target cells, exerting proliferative, cytostatic, chemoattractant, or differentiative effects. Most cytokines have biological activities in the high pM concentration range and are coupled to intracellular signaling and second-messenger pathways via specific high-affinity recep-

tors on target cell membranes. The cytokine and growth factor nomenclature reflects the historical description of biological activity rather than a systematic relationship based on molecular structure.

RANTES

RANTES (Regulated on Activation, Normal T-Cell Expressed and Secreted) is a cytokine of the β or "C-C" chemokine family. RANTES is a chemoattractant for monocytes and memory T-cells. Although the original studies identified RANTES secretion by hematopoietic cells, this cytokine also has been shown to be secreted by some epithelial and mesenchymal cells. RANTES may be an important mediator in both acute and chronic inflammation. Numerous potential binding sites for transcription factors regulate this gene, and both early- and late-activating transcription factor complexes have been described (40).

In normal endometrium, RANTES protein immunolocalization predominates in the stromal compartment (41). In vitro, stromal-cell cultures synthesize RANTES mRNA and secrete protein when induced by TNF- α and IFN- γ stimulation, whereas epithelial cells synthesize neither transcripts nor protein (41). The discrepancy between basal RANTES expression in vivo and in vitro, and the requirement of cytokine stimulation in cultured cells for expression of RANTES, may indicate that cytokines from resident immune cells stimulate paracrine tissue effects in vivo. In cultures of purified cells, TNF- α must be added exogenously to effect significant RANTES gene expression (41).

In endometriotic lesions, the pattern of RANTES protein distribution was similar to that found in a normal endometrium (41). However, an important distinction between normal endometrial stromal-cell cultures and those derived from endometriomas is that under similar conditions, endometrioma-derived stromal-cell cultures secrete significantly greater concentrations of RANTES (42). In this way, peritoneal implants may contribute to the increased pelvic fluid concentrations of RANTES seen in patients with moderate and severe endometriosis (39).

We have proposed that secretion of RANTES by ectopic implants provides a mechanism for peritoneal leukocyte recruitment. Expression of the RANTES gene is up-regulated in endometriotic stromal cells in response to IL-1 β , a pleiotropic, macrophage-derived cytokine. These findings are consistent with a feed-forward inflammatory loop whereby IL-1 β from activated macrophages can lead to RANTES production and further monocyte chemotaxis (unpublished observations). We postulate that this pathway is activated in endometriosis.

Interleukin-1

The cytokine interleukin-1 (IL-1) plays a central role in the regulation of inflammation and immune responses. Originally recognized as a product of activated monocytes and macrophages, IL-1 now is known to affect the activation of

T-lymphocytes and the differentiation of B-lymphocytes, as well (Fig. 2). Two receptor agonists, IL-1 α and IL-1 β , are encoded by different genes, and although the two proteins share only 18%–26% amino acid homology, they bind to the same receptors and have comparable biological activities. A related protein, the IL-1 receptor antagonist (IL-1ra), is an endogenous inhibitor that blocks the binding of both IL-1 α and β to IL-1 receptor type I.

The critical importance of the IL-1 system was suggested by an experiment in which successful implantation in mice was blocked with the administration of exogenous IL-1ra (43). However, IL-1 gene knock-out experiments do not corroborate this observation (44). IL-1 has been isolated from the peritoneal fluid of patients with endometriosis. Results have been inconsistent, with some investigators demonstrating elevated concentrations in patients with endometriosis (36, 45, 46) and others finding no elevation (47).

Mori et al. (48) showed increased levels of IL-1 β mRNA expression in peritoneal macrophages from patients with mild endometriosis, but increased levels of IL-1ra mRNA expression in macrophages from patients with moderate to severe endometriosis. The coordinated effects of these proteins are not yet clear.

A role for IL-1 β in promoting angiogenesis in endometriotic lesions has been suggested by the induction of angiogenic factors (vascular endothelial growth factor and interleukin-6) in endometriotic stromal cells but not in normal endometrial stromal cells (49). Viganò et al. have suggested yet another feature of IL-1 β in the pathophysiology of endometriosis. These investigators have found that IL-1 β increased sICAM-1 shedding from endometrial cells. As explained above, sICAM-1 protein may interfere with peritoneal immunosurveillance and thus allow refluxed endometrial tissue to escape destruction in the peritoneal cavity (19).

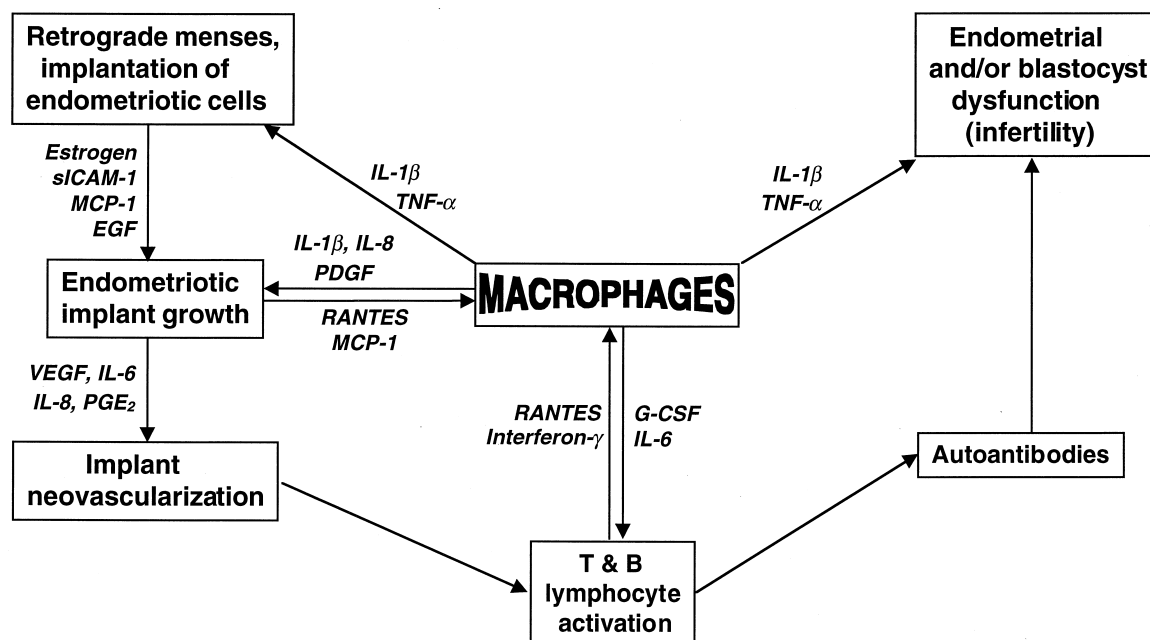
Interleukin-6

The pleiotropic cytokine interleukin-6 (IL-6) is an important regulator of inflammation and immunity that serves as a physiologic link between the endocrine and the immune systems. Like IL-1, IL-6 modulates secretion of other cytokines, promotes T-cell activation and B-cell differentiation, and inhibits growth of various human cell lines. It is a 23–26 kDa phosphoglycoprotein that exists in several isoforms and is produced by a number of cell types, including monocytes, macrophages, fibroblasts, endothelial cells, vascular smooth-muscle cells, and endometrial epithelial and stromal cells (50, 51). In addition, IL-6 is produced by several endocrine glands, including the pituitary and the pancreas.

Endometrial stromal and epithelial cells produce IL-6 in response to hormones and other immunologic activators. Endometrial stromal-cell IL-6 protein is induced by IL-1 α or β , tumor necrosis factor, platelet-derived growth factor, and interferon- γ (49, 51). Because estrogen typically enhances proliferation in the endometrial epithelium, it is proposed

FIGURE 2

Macrophages play a central role in the immunobiology of endometriosis. Interactions between the endometriotic cells and macrophages are mediated by growth factors, cytokines, and chemokines, leading to the paradoxical survival of implants rather than their demise.



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that estrogen causes epithelial cell proliferation by limiting the synthesis of epithelial cell inhibitors such as IL-6.

Zarmakoupis et al. have found that IL-6 inhibits the proliferation of human endometrial stromal cells, and that this growth inhibition is dependent on cell density, suggesting that IL-6 may play a role in epithelial–stromal interactions governing the regulation of normal uterine function (52). Tabibzadeh et al. (50) have suggested that IL-6 fluctuations during the menstrual cycle reflect an inverse relationship to estrogen action; estrogen concentrations are high in the proliferative phase of the cycle, while IL-6 levels are low. Conversely, unopposed estrogenic activity is low during the secretory phase of the cycle, whereas IL-6 levels are high.

In vitro studies suggest a dysregulation of IL-6 response in the peritoneal macrophages (53), endometrial stromal cells (54), and peripheral blood macrophages (55) of patients with endometriosis. As proposed by Rier et al., endometriotic implants may be resistant to growth inhibition by IL-6 because expression of cell surface IL-6R is decreased. This decrease may be the reason for lower concentrations of IL-6 soluble receptor (IL-6sR) in the peritoneal fluid of these patients (56).

It should be noted, however, that other investigators have

reported an increased concentration of IL-6sR in the peritoneal fluid of patients with endometriosis (57). Furthermore, conflicting results have been reported on the level of immunoreactive IL-6 detected in the peritoneal fluid of patients with endometriosis; some investigators have demonstrated elevated concentrations (56, 58), whereas others have found no elevation (59, 60). Our studies in peritoneal fluid failed to demonstrate statistically significant differences in IL-6 levels between controls and endometriosis patients (13). These inconsistent findings likely are related to antibody specificity.

Tumor Necrosis Factors

The tumor necrosis factors (TNFs) are pleiotropic cytokines with a range of beneficial and injurious effects, depending on the quantity produced, their tissue localization, the local activity of TNF-binding proteins, and their hormonal and cytokine milieu.

TNF- α is produced by neutrophils, activated lymphocytes, macrophages, NK cells, and several nonhematopoietic cells, whereas TNF- β is produced by lymphocytes. Although these TNFs were initially identified for their ability to kill certain cell lines, their primary function is their ability (along with IL-1) to initiate the cascade of cytokines and other factors associated with inflammatory responses. In contrast

to their similar biological activities, regulation of the expression and processing of these two factors is quite different.

In the human endometrium, TNF- α has been implicated in the normal physiology of endometrial proliferation and shedding. The greatest expression of protein and message is in the epithelial cells, with most staining occurring during the secretory phase (61). Staining of stromal cells is also detected, predominantly in the proliferative phase of the cycle, suggesting differential local and hormonal regulation of this cytokine (62). Short-term explant cultures secrete immunoreactive TNF- α protein (63).

In vitro, epithelial cell TNF- α secretion is modulated by IL-1, progesterone, and PP14 (64). TNF- α has been shown to increase the amount of prostaglandin produced by cultured endometrial epithelial cells (65), and to promote the adherence of cultured stromal cells to mesothelial cells (66). This latter finding suggests that the presence of TNF- α in pelvic fluid may promote adherence of ectopic endometrial tissue to the peritoneum, allowing implants to form.

Several investigators have shown that TNF- α concentrations are elevated in the peritoneal fluid of patients with endometriosis, and that these higher concentrations correlate with the stage of the disease (67). The source of the elevated TNF- α concentration in the pelvic fluid of endometriosis patients is not clear, but in vitro studies suggest that peritoneal macrophages (47) and peripheral blood monocytes (55) from these patients also have up-regulated TNF- α protein secretion. Because of its importance in other inflammatory processes, it is likely that this cytokine plays a role in the pathogenesis of endometriosis.

Vascular Endothelial Growth Factor

Studies from our laboratory and others have focused on the proliferation and neovascularization of the nascent endometriosis implant. The reader is referred to recent reviews on this subject (13, 68, 69).

Vascular endothelial growth factor (VEGF), also known as vascular permeability factor and vasculotropin, is one of the most potent and specific angiogenic factors. The biochemistry and molecular biology of VEGF are well-described (70). VEGF binds to a family of tyrosine kinase receptors including Flt-1 and KDR, leading to dimer formation, autophosphorylation of the receptor, and, as with FGF, activation of mitogen-activated protein kinases (71). Although the precise signaling pathway for VEGF remains unknown, VEGF-receptor activation leads to a rapid increase in intracellular Ca^{2+} and inositol triphosphate concentrations in endothelial cells.

The regulation of bioavailable VEGF is controlled at the transcriptional and post-translational levels. Four distinct mRNA species arising via differential splicing of a primary VEGF transcript have been identified and characterized. The most common VEGF transcripts encode proteins comprising 165 and 121 amino acid residues that form glycosylated

homodimers of about 45 and 35 kDa, respectively. Longer forms of VEGF (189 and 206), despite containing identical hydrophobic signal sequences, are not actively secreted into the extracellular milieu (70). Instead, the relatively basic carboxyl-termini of these isoforms cause them to become reversibly associated with heparan-sulfate proteoglycans of the extracellular matrix. These bound isoforms may have juxtacrine effects. Release of bioactive VEGF fragments from this extracellular reservoir can be effected by heparin, hypoxia, or plasmin cleavage (72).

In the endometrium, the 165 and 121 variants appear to be the predominant isoforms (73, 74); however, only semiquantitative RT-PCR methods were used. A novel endometrial mRNA transcript predicting a 145-residue protein was reported by Charnock-Jones et al. (73), but that finding was not confirmed by Torry et al. (75), and no protein corresponding to this transcript has been demonstrated.

VEGF mRNA was detected in normal endometrium by RNase protection analyses, and its expression was found to be highest during the secretory phase of the menstrual cycle (75, 76). VEGF protein was localized predominantly in endometrial glands, and stromal staining was less abundant and more diffuse (76, 77). Expression of the VEGF gene in normal human endometrial (76) and adenocarcinoma cells (73) is acutely up-regulated by estradiol in vitro, as it is in the rodent uterus in vivo (78). Other factors known to up-regulate VEGF expression include hypoxia, IL-1 β , PDGF and TGF- β , EGF, and PGE₂ (49, 79, 80).

VEGF-induced angiogenesis allows repair of the endometrium following menstruation. It also appears to increase microvascular permeability, permitting the formation of a fibrin matrix for endothelial migration and proliferation (81). As a result, increased fluid and protein extravasation cause local edema which help to prepare the endometrium for embryo implantation (76, 82).

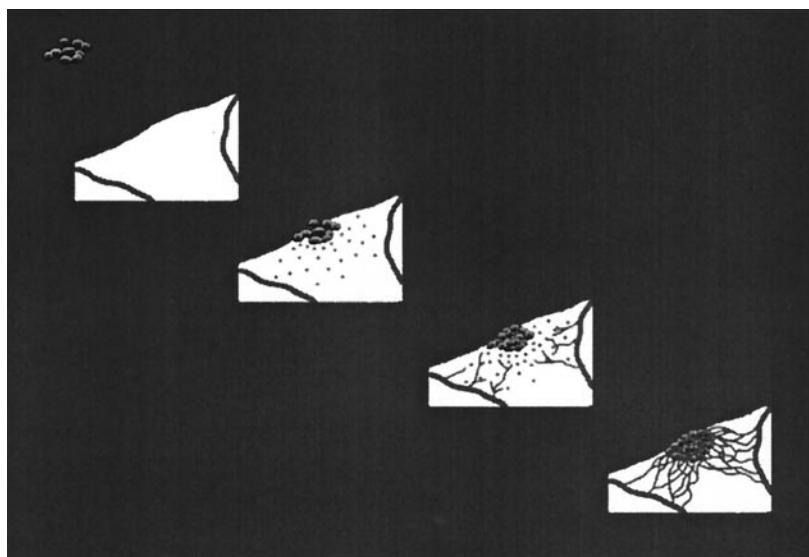
The expression of VEGF by endometriotic implants provides a mechanism for the neovascularization that is commonly observed around these lesions (83) (Fig. 3). VEGF immunostaining was observed in the epithelium of endometriotic implants (76), particularly in hemorrhagic red implants (68), and increased concentrations of soluble VEGF in pelvic fluid from patients with endometriosis have been reported. The cellular sources of VEGF in peritoneal fluid have not been precisely defined. Although evidence exists to suggest that endometriotic lesions produce this factor (49, 76), activated peritoneal macrophages also have the capacity to synthesize and secrete VEGF (84).

Anti-Inflammatory Agents in the Treatment of Endometriosis

If, as proposed here, the signs and symptoms of endometriosis reflect the sequelae of inflammatory reactions at the sites of ectopic implants, anti-inflammatory agents should be

FIGURE 3

Schematic illustration of the theorized genesis of endometriotic lesions and their vascular supply. Endometrial cells alight onto the peritoneum, where they secrete vasculogenic proteins into the subperitoneal space. Angiogenesis accounts for the lesions' obtaining an independent vascular network, which permits their continued survival and growth.



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clinically useful as therapy. Evidence supports the use of cyclo-oxygenase inhibitors for the treatment of pain and dysmenorrhea associated with endometriosis (85), but therapeutic benefits of these agents on pregnancy outcome have not been documented.

The androgen danazol has been used extensively for the treatment of endometriosis. Although it was initially thought to act as a selective antigonadotropin (86), more recent studies indicate that danazol has multiple effects. Using an *in vitro* mixed lymphocyte culture assay, Hill and colleagues showed that this agent manifested immunosuppressive effects similar to those observed with dexamethasone and high-dose progesterone (87).

One of the anti-inflammatory agents that has been investigated is the phosphodiesterase inhibitor pentoxifylline (88). In the early 1990s, two papers published by Steinleitner and collaborators provided dramatic evidence that pentoxifylline might fully reverse the infertility observed in rodent models of endometriosis. When syngeneic mice received heterologous activated macrophages, fertilization occurred in <1%. However, periovulatory pretreatment with pentoxifylline reversed this inhibition by >99% (89). Using a Golden hamster model, the same group of investigators developed a surgical model of endometriosis by implanting uterine fragments into the uterine mesentery (90). They observed a 98% decrease in fertility, which was completely reversed when the animals were treated with periovulatory injections of

pentoxifylline (2.5 mg/kg/8 hours). Attempts to confirm the former study were unsuccessful (91).

In a surgically induced model of endometriosis in the rat, Nothnick et al. (92) showed that implant surface area was reduced by 88% in animals receiving pentoxifylline. A randomized, 12-month, placebo-controlled clinical trial of oral pentoxifylline (800 mg p.o./day) was undertaken recently (93). Life-table analysis in 60 subjects demonstrated a 31.0% overall pregnancy rate in the pentoxifylline group and an 18.5% rate in the placebo group; however, this increase did not achieve statistical significance (95% CI = 0.18–1.67).

Incorporating the Immune Response into a New Classification of Endometriosis

The current revised classification of endometriosis put forth by the American Society for Reproductive Medicine (ASRM) in 1985 (94) poorly prognosticates differences in pregnancy rates among the four classification groups (95). A valuable classification scheme should forecast the likelihood of response in symptomatology and fertility to various therapies. We believe a useful endometriosis classification scheme should incorporate the immunobiological milieu that the present classification fails to take into account. Using some of the aforementioned biomarkers of inflammation (96), it might be possible to establish a classification scoring system that more accurately defines those endometriosis

patients who will most likely benefit from specific therapies—with respect to both pain and fertility.

SUMMARY

Clinical studies supporting a direct relationship between lesion volume and the pain (97, 98) or infertility (99, 100) symptoms of endometriosis remain controversial. Recent evidence suggesting that the symptoms associated with endometriosis result from local peritoneal inflammatory reactions may explain this conundrum.

Increased concentrations of activated pelvic macrophages and lymphocytes, as well as the elevated levels of specific cytokines and growth factors reviewed above, support this hypothesis. The precise roles of these soluble factors are unknown currently, but we propose that a complex network of locally produced cytokines modulate the growth and inflammatory behavior of ectopic endometrial implants. The secretion of proinflammatory and mitogenic proteins by endometriotic lesions and associated immune cells into the peritoneal microenvironment contributes to a cascade of events that result in implant proliferation and invasion, the recruitment of capillaries to the growing lesions, and further chemoattraction of leukocytes to these foci of peritoneal inflammation.

Future therapeutic strategies to ameliorate the inflammatory reaction associated with endometriosis should target these potentially pathological proteins, but we cannot ignore the likely physiological actions of many of the same bioactive molecules in normal eutopic endometrial function.

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